Kernel, Frame, Winding, Residue

A Geometric Classification of Binary Digit-Extraction Formulas,

with BBP as the Canonical Representative of an Equivalence Class

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Abstract

The Bailey-Borwein-Plouffe formula is normally treated as a discovered integer relation whose parameters -- shell spacing 8, base 16, coefficient vector {4, -2, -1, -1} -- are facts rather than consequences. This paper derives all of them, shows that BBP is one member of a two-family class rather than an isolated result, and then establishes that the coefficient vector is not a solution but a canonical representative of an infinite equivalence class.

The derivation begins at the kernel. The denominator 1 - x^n is the closure condition x^n = 1, whose zero set is the n-th roots of unity; the frame is therefore the pole structure of the formula itself, not an interpretation applied afterward. Under the n-th root of unity the exponent decomposition nk + r becomes the quotient map from the integers onto the integers modulo n: complete windings vanish and only the residue survives, as an orientation.

Digit extraction requires an endpoint that is simultaneously a mark coordinate of the frame and a half-integer power of the radix. Exhaustive sweep over every regular frame with integer degree measure and every integer base from 2 to 20 yields exactly two admissible angles, 45 and 60 degrees -- the two special right triangles. These generate an octal family (8 divides n, base 2^(n/2)) and a hexagonal family (6 divides n, base 2^n). Integer relation detection confirms both: the octal frame returns BBP exactly, and the hexagonal frame, previously unexamined, returns two relations for 4*pi*sqrt(3) and 4*pi/sqrt(3) over denominators 6k+r at

base 64, verified to fifty decimal places. The constant each frame produces inherits the tangent of its characteristic angle.

Two growth operators are distinguished. Refinement doubles the frame and is destroyed at the first step, because the half-angle formula introduces square roots that are not powers of the radix; convergence is polynomial. Winding leaves the frame fixed and adds one traversal, contributing exactly one power of the base; compatibility is preserved indefinitely and convergence is geometric. The frame is complete at the first winding: growth occurs in the weight, never in the geometry.

Finally, the status of the coefficient vector is settled. The constraint passes the derivative screen but has rank one on eight unknowns, leaving a seven-dimensional surface. Integer relation detection on the shell constants alone, with no target, returns an explicit null vector, verified to 10^-80, along which the observable is exactly invariant. Consequently pi admits infinitely many integer representations in the same frame, and the invariant object is the equivalence class rather than the vector. Rank does not select; integrality does not select; minimality does, and it is shown to be a genuine rank-reducing constraint because its gradient is not orthogonal to the null direction. BBP is the shortest vector in its class, with squared norm 22 against 135 and 343 for its neighbours. The search landscape is exactly flat along the null direction, which is why the discovery tool must be lattice reduction rather than descent.

§1 Definitions

Definition 1 (Frame). A frame of order n is the cyclic set of directions generated by the n-th roots of unity, F_n = { exp(2*pi*i*j/n) : 0 <= j < n }. Each residue class modulo n indexes exactly one element. Elements are called marks; a mark at angle theta has coordinate cos(theta).

Definition 2 (Kernel). A kernel of order n is the rational function with denominator 1 - x^n. Its poles are exactly the elements of F_n.

Definition 3 (Shell). For a kernel of order n, the shell of index (k, r) is the denominator nk + r in the expansion of the kernel about the origin, with k a non-negative integer and r in {1, ..., n}.

Definition 4 (Endpoint). The endpoint u is the upper limit of integration. It is frame-compatible if u = cos(theta) for some mark of F_n, and radix-compatible if u^n = 1/b for an integer base b.

Definition 5 (Winding, orientation). In nk + r, the quotient k is the winding count and the residue r is the orientation.

Definition 6 (Shell constant). B_r = SUM over k of 1/(b^k * (n*k + r)). These are the basis constants of the frame; each is the accumulated contribution of one marked direction.

§2 The Kernel Frame Theorem

Theorem 1. *The frame is not chosen. It is the zero set of the kernel denominator.*

Proof. Setting 1 - x^n = 0 gives x^n = 1, whose solutions are the n-th roots of unity, which is Definition 1. Square.

For BBP the denominator is 1 - x^8, so n = 8 immediately, with no external geometric assumption. The geometry is not an interpretation laid over an analytic identity; it is the singularity structure of the identity. The denominators 8k + r are one per marked direction, indexed by winding.

§3 The Projection Theorem

Theorem 2. *Under repeated application of the n-th root of unity the winding count is annihilated and only the orientation survives.*

Proof. With omega = exp(2*pi*i/n), omega^(nk+r) = (omega^n)^k * omega^r = omega^r. The map nk + r maps to r is the quotient homomorphism from the integers onto the integers modulo n. Square.

Verified for n = 8, r = 1, k = 0 through 6: shell indices 1, 9, 17, 25, 33, 41, 49 are seven distinct magnitudes; the angle of omega raised to each is 45.00 degrees in every case.

Remark. The magnitude is not destroyed. The map is a quotient and k remains recoverable from the integer; it is projected out by a choice of representation, not lost. Likewise directions do not rotate -- operators do. The integers index applications of the rotation operator; they carry operator depth, not quantity.

§4 Binary Compatibility and the Two Families

Theorem 3. *Within the class of marks of regular frames with integer degree measure, and integer bases, exactly two angles are simultaneously frame- and radix-compatible: 45 degrees and 60 degrees.*

Evidence. Exhaustive sweep over every regular frame of order n with n dividing 360, every mark strictly between 0 and 90 degrees, and every integer base from 2 to 20.

angle	cosine	equals	triangle
45 degrees	0.7071067812	2^(-1/2)	45-45-90 (1, 1, sqrt2)
60 degrees	0.5000000000	2^(-1)	30-60-90 (1, sqrt3, 2)

Base 2 returns both; base 4, being a power of 2, returns 60 degrees; every other integer base from 3 to 20 returns nothing. Changing the radix does not open new frames, because the binding constraint is that the cosine be a pure power, and only these two rational-degree cosines are.

family	condition	endpoint	base	first instances
OCTAL	8 divides n	cos 45 = 2^(-1/2)	2^(n/2)	n=8 -> 16, n=16 -> 256, n=24 -> 4096
HEXAGONAL	6 divides n	cos 60 = 2^(-1)	2^n	n=6 -> 64, n=12 -> 4096, n=18 -> 262144

Scope. This is an exhaustive finite sweep over a stated class, not a theorem for all real angles. A complete result would require determining, from the theory of cyclotomic fields, which rational-degree cosines lie in the multiplicative group generated by 2. See Omega 1.

Retraction. A previous version reported a null result for "base 8 with 6 sectors." That test used the endpoint $2^{-1/2}$, which is not a mark coordinate of the hexagonal frame, whose coordinates are -1, -0.5, 0.5 and 1. The endpoint did not lie on its own frame. The correct pairing for n = 6 is base 64. The earlier null is withdrawn as a failed embedding, not a failed family -- and the corrected pairing is where the new result of Section 8 was found.

§5 Minimality of the Octal Frame

Theorem 4A (Angular). *45 degrees is a mark of F_n iff 8 divides n; least such n is 8.* The marks lie at multiples of 360/n, and 45 is such a multiple iff 360/n divides 45. Square.

Theorem 4B (Endpoint). *$2^{-1/2}$ is a mark coordinate of F_n iff 8 divides n; least such n is 8.* Coordinate sets: n = 2 gives {-1, 1}; n = 4 gives {-1, 0, 1}; n = 6 gives {-1, -0.5, 0.5, 1}; none contains 0.7071. n = 8 contains plus and minus 0.707107. Square.

Corollary. Both require 8 to divide n; with $b = 2^{n/2}$ the minimum is n = 8, b = 16. Base 16 is the smallest base in which 45 degrees exists as a direction. That is why the formula is hexadecimal: not hardware convenience, but the minimum frame in which the target angle has a name.

§6 The Weights, Derived

Lemma 1. *INTEGRAL from 0 to u of $x^p/(1 - x^n)$ dx = u^{p+1} * SUM over k of $(u^n)^k / (n*k + p + 1)$.* Expand the kernel as a geometric series and integrate term by term. Square.

Corollary 1 (Shell index). *r = p + 1.* Integrating x^p produces $x^{p+1}/(p+1)$. Mechanical, and it explains why the BBP numerator support {0, 3, 4, 5} corresponds to shells {1, 4, 5, 6} without exception.

Corollary 2 (Weight formula). *$w_r = a_p * u^r$.* For the octal frame $u^r = 2^{-r/2}$.

power p	coefficient a_p	shell r	factor 2^(-r/2)	weight
0	4*sqrt2 = 5.65685425	1	0.707106781	4
3	-8	4	0.250000000	-2

power p	coefficient a_p	shell r	factor 2^(-r/2)	weight
4	-4*sqrt2 = -5.65685425	5	0.176776695	-1
5	-8	6	0.125000000	-1

The recovered vector is {4, -2, -1, -1}, and the weighted sum agrees with pi to 25 decimal places.

Lemma 2 (Parity). *For u = 2^(-1/2), the factor u^r is rational when r is even and a rational multiple of sqrt(2) when r is odd.* Since the weights are integers, the numerator coefficient must carry a compensating sqrt(2) precisely when r is odd. The placement is forced, not stylistic. And by Theorem 2, odd r are the diagonal directions -- so sqrt(2), the reciprocal of the diagonal coordinate, appears in the algebra exactly at the diagonal terms.

§7 Orbit Topology and the Angular Contribution

The four surviving shells split two and two under two equivalent descriptions. Arithmetically, odd r gives odd moduli coprime to 16, so 16 is invertible and its orbit returns to 1; even r gives even moduli sharing a factor with 16 (4 for r = 4, 2 for r = 6), so 16 is not invertible and the orbit falls into a short trapped cycle. Geometrically, odd r are diagonals and even r are axes. The chain is parity to invertibility to orbit topology.

```
90   (r=2, axis)
|
135  (r=3)   |   45  (r=1)  DIAGONAL, ROTOR, w=+4
\    |    /
180 -----------+---+---+----------- 0   (r=8, axis)
(r=4, AXIS,   /   |   \
DAMPER,   /    |    \
w=-2)  225  (r=5) |  315  (r=7)
DIAGONAL,    |
ROTOR, w=-1  270   (r=6, AXIS, DAMPER, w=-1)
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Figure 1. The octal frame. Residues 1 and 5 are 180 degrees apart: one line, the diagonal, read from both ends and weighted +4 and -1. Residues 4 and 6 are the negative axes. The formula is

one diagonal plus two axes. The endpoint is the diagonal coordinate, so integration proceeds along the rotor line and the axis terms are what it is subtracted against.

Theorem 5 (Angular Contribution). *Poles on the real axis contribute logarithms of magnitudes; poles off it contribute arguments.* In partial fractions each term integrates to R_j * log(x - omega_j), and log(z) = log|z| + i*arg(z): the real part is a length, the imaginary part an angle. A pole on the real axis has no argument to give. Since pi = 4*arctan(1), reaching it requires a pole at 45 degrees. Consistent with Section 8, where log 3 appears on the axis residues 2 and 6 while pi appears on residues including both diagonals. **Status: interpretive.**

§8 Growth Operators and the Classification

Definition 7. Refinement is F_n maps to F_2n: more directions, same traversals. Winding is k maps to k+1: same directions, one further traversal.

Theorem 6. *Refinement removes radix compatibility at the first step, in both families.* The half-angle formula gives cos(theta/2) = sqrt((1 + cos theta)/2). From cos 60 = 2^(-1) it gives cos 30 = 0.8660254038; from cos 45 = 2^(-1/2) it gives cos 22.5 = 0.9238795325. Neither is a power of 2. Square.

Theorem 7. *Winding preserves radix compatibility indefinitely.* By Lemma 1 the contribution of winding k is (u^n)^k = 1/b^k, a power of the base at every step. No new algebraic number enters. Square.

step	REFINE frame	value	error	WIND k	value	error
1	n=6	3.000000000000	1.42e-01	1	3.133333333333	8.26e-03
5	n=96	3.141031950891	5.61e-04	5	3.141592645460	8.13e-09
10	n=3072	3.141592106043	5.48e-07	10	3.141592653590	1.78e-15

Refinement divides the error by about 4 per step, polynomial; winding by about 16, geometric. After ten steps they differ by nine orders of magnitude, and only winding retains random access.

Corollary (Frame completion). At k = 0 the shells n*0 + r for r = 1 through n already cover every mark. The geometry is complete at the first winding; every subsequent k is the same marks at 1/b the weight, and the winding projects to unity by Theorem 2. Nothing is constructed; the aggregate reading sharpens. This inverts the usual convergence model: rather than more steps producing better structure, complete structure permits better measurement.

§8.1 The hexagonal family carries its own identities

Integer relation detection at 60-digit precision. Octal control (n = 8, base 16) recovers BBP exactly, and additionally returns relations for log 2, log 3 and log 5, with log 3 supported only on the axis residues 2 and 6. Hexagonal (n = 6, base 64), previously unexamined, returns two relations, verified to 50 decimal places after clearing denominators:

*4*pi*sqrt(3) = 20/(6k+1) + 6/(6k+2) - 1/(6k+3) - 3/(6k+4) - 1/(6k+5), over 64^k*

*4*pi/sqrt(3) = 12/(6k+1) - 6/(6k+2) - 3/(6k+3) - 3/(6k+4), over 64^k*

relation	frame	base	series value	|error|
4*pi*sqrt3	n=6	64	21.765592370810614 2071	4.3e-50
4*pi/sqrt3	n=6	64	7.2551974569368714 0238	1.1e-50
pi (BBP control)	n=8	16	3.1415926535897932 3846	3.2e-50

The constant inherits the frame tangent. tan 45 = 1, so arctan(1) = pi/4 and the octal frame gives clean pi. tan 60 = sqrt(3), so arctan(sqrt(3)) = pi/3 and the hexagonal constants carry sqrt(3). The square root appearing in each family is the tangent of its own characteristic direction. BBP is the octal member of a frame-winding class, not a singular formula.

§9 The Null Lattice: What the Coefficient Vector Actually Is

This section replaces a note in the previous version which stated that the BBP vector is "minimal, not unique." That is correct and under-specified. The situation is sharper and stranger.

§9.1 The derivative screen and the rank count

A constraint C(y) = 0 can determine y only if it contains y: if the partial derivative of C with respect to every component of y vanishes, the constraint lives in another sector and cannot select, however deep it is. For the coefficient problem, C(w) = SUM over r of w_r B_r - pi, and:

dC/dw_r = B_r

All eight shell constants are nonzero ($B_1 = 1.00718$, $B_2 = 0.50648$, ..., $B_8 = 0.12908$). The constraint is therefore coupling-type and eligible. But the Jacobian is a single row: rank 1 on 8 unknowns, leaving a seven-dimensional solution surface. Rank does not select the BBP vector; it only reports that the surface is large.

§9.2 The null vector

Integer relation detection was run on the eight shell constants alone, with no target. It returns:

$$-8 B_1 + 8 B_2 + 4 B_3 + 8 B_4 + 2 B_5 + 2 B_6 - 1 B_7 + 0 B_8 = 0$$

verified to 4.9×10^{-80} at 80-digit working precision. It uses seven of the eight shells -- every direction except r = 8, the positive x-axis. Consequently pi has infinitely many integer representations in the same frame:

t	weight vector w + t*null	value	equals pi
0	[4, 0, 0, -2, -1, -1, 0, 0]	3.14159265358979324	yes
1	[-4, 8, 4, 6, 1, 1, -1, 0]	3.14159265358979324	yes
-1	[12, -8, -4, -10, -3, -3, 1, 0]	3.14159265358979324	yes
2	[-12, 16, 8, 14, 3, 3, -2, 0]	3.14159265358979324	yes

Searching for further independent null directions by dropping each coordinate in turn returned none, so the null lattice has rank 1 within the range examined.

§9.3 The null direction is a latent distinction

Each row above is a genuinely different weight vector -- a real distinction in the representation -- producing an identical observable. The frame has no operation that separates them. This is precisely the structure of a latent distinction: two states differ, and no admissible map reveals the difference, so the distinction is structurally present and computationally inert.

In tangent-space terms, the null vector satisfies grad C . null = 0, so it lies in T = { v : grad C . v = 0 }, the space of motions the constraint permits. It is an integer point in the tangent space of the constraint surface.

Consequence. The weight vector is not the invariant. It is a coordinate, defined only modulo the null lattice L. The invariant object is the equivalence class [w] in Z^8 / L; the observable pi is constant on the entire class. Any representative computes pi correctly, and BBP is one of them.

§9.4 Minimality is the selector, and it is a genuine cut

Rank leaves seven dimensions. Integrality cuts the surface to a lattice but, by Section 9.2, that lattice is infinite along L. Neither selects. The remaining candidate is minimality, and the test for whether a second constraint genuinely removes a degree of freedom is whether its gradient fails to be orthogonal to the tangent direction. For C_2(w) = ||w||^2:

*grad C_2 . null = 2 * (w . null) = 2 * (-52) = -104   (nonzero)*

Minimality therefore cuts the tangent direction and is a genuine rank-reducing constraint. The real minimiser along the null line sits at t = 52/217 = 0.2396, whose nearest integer is 0. Verifying by squared norm:

t	w + t*null	squared norm
-2	[20, -16, -8, -18, -5, -5, 2, 0]	1098
-1	[12, -8, -4, -10, -3, -3, 1, 0]	343
0	[4, 0, 0, -2, -1, -1, 0, 0]	22
1	[-4, 8, 4, 6, 1, 1, -1, 0]	135
2	[-12, 16, 8, 14, 3, 3, -2, 0]	682

The BBP vector is the shortest vector in its class. It is the canonical representative, not the unique solution. The selection cascade is: derivative screen (eligible), rank (7-dimensional surface, no selection), integrality (infinite lattice, no selection), minimality (selects). Only the last does any work.

§9.5 Why the discovery tool is lattice reduction and not descent

Along the null direction the observable does not change at any distance: at t = 0, 1, 5 and 50 the value is 3.141592653589793238463 in every case, with deviations at the 10^-70 level attributable to working precision. The landscape is exactly flat throughout the tangent space. No local information anywhere on the surface points toward the short vector.

This is the same distinction that separates search problems whose solutions are isolated points with no local slope from those whose predicates are clustered and coupled, permitting descent. The BBP coefficient problem is of the flat kind: one cannot descend to the answer, one must reduce. The geometry of the search space, not the difficulty of the arithmetic, is what dictates that the tool be PSLQ. And since PSLQ returns the shortest relation by construction, "minimal" and "what PSLQ found" are the same statement.

§9.6 A correction, and what it reveals

An intermediate result appeared to show a genuinely different representation of pi over sixteen shells at base 256. It is not. The shells carrying nonzero weight are 1, 4, 5, 6, 9, 12, 13, 14, and 9, 12, 13, 14 are exactly 8 + {1, 4, 5, 6}: the second winding of the same four directions. Their weights are the BBP weights divided by 16 -- one winding of decay. Since base 256 = 16^2, one step in the sixteen-frame is two steps in the eight-frame, so the vector is BBP with two windings written out longhand. The claim of a new representation is retracted. What it actually shows is Theorem 7 appearing explicitly inside a coefficient vector.

§9.7 Structure precedes the interface

One may ask whether the admissible structure exists prior to any procedure that reads it, or whether the reading generates it. For this domain the question is decidable. The null lattice L is a property of the shell constants alone; it is determined by the frame and the base, and exists whether or not any relation-detection algorithm is ever executed. PSLQ selects a representative from the class but does not create the class. The structure is prior; the interface samples it.

Restated in channel terms: the weight vector is representation-dependent, free along L, and invisible to the observable -- it is shape. Pi is constant across the entire class -- it is value. The null direction is pure shape carrying zero value: the frame's own blind spot, of dimension at least one.

§10 Epistemic Status

Claims are typed by what they contribute. A Class I statement introduces a coordinate relation without reducing the admissible set. Class II removes a boundary without reducing dimension. Class III removes a degree of freedom. Only Class III contributes rank.

Claim	**Class**	**Status**
Frame = kernel zero set (Thm 1)	I	[V] proved
Winding annihilated by projection (Thm 2)	I	[V] proved, verified k=0..6
Only 45 and 60 degrees admissible (Thm 3)	III	[M] exhaustive over stated class

Claim	Class	Status
Angular minimality, 45 first at n=8 (4A)	III	[V] proved
Endpoint minimality, cos45 first at n=8 (4B)	III	[V] proved
r = p+1 (Cor 1)	I	[V] proved
w_r = a_p u^r giving {4,-2,-1,-1} (Cor 2)	I	[V] computed, 25 digits
Parity Lemma: sqrt2 on odd shells only (Lem 2)	II	[V] proved
Refinement destroys compatibility (Thm 6)	III	[V] proved, both families
Winding preserves it (Thm 7)	III	[V] proved
Hexagonal relations, 4*pi*sqrt3 and 4*pi/sqrt3	III	[V] PSLQ, verified 50 digits
Constraint is coupling-type (derivative screen)	II	[V] all B_r nonzero
Null vector exists, rank 1 in range examined	III	[V] verified to 1e-80
Invariant is the class, not the vector	I	[V] follows from the null vector
Minimality cuts the tangent direction	III	[V] grad dot null = -104
BBP is the shortest vector in its class	III	[V] norms 22 vs 135, 343
Angular Contribution (Thm 5)	I	[I] interpretive
Base-8-with-6-sectors null (v1)	--	[R] retracted, malformed test
16-shell "new representation" (intermediate)	--	[R] retracted, BBP unrolled

[V] verified by proof or computation. [M] measured exhaustively over a stated finite class. [I] interpretive. [R] retracted.

§11 Open Problems

Omega 1 (General binary compatibility). Theorem 3 is an exhaustive sweep over a finite class. A complete result would use cyclotomic field theory to determine which rational-degree cosines lie in the multiplicative group generated by 2. Conjecture: only cos 45 and cos 60.

Omega 2 (Rank of the null lattice). One null direction was found, and the coordinate-dropping search returned no second independent one within the coefficient bound used. Whether the null lattice of F_8 has rank exactly 1, and what its rank is for general n, is open. The rank determines the dimension of the equivalence class and therefore how much freedom minimality must remove.

Omega 3 (Higher members). The octal family continues at n = 16 (base 256) and n = 24 (base 4096); the hexagonal at n = 12 (base 4096) and n = 18. Whether these carry relations not reducible to unrolled windings of the minimal members, by the mechanism of Section 9.6, is unexamined.

Omega 4 (Crossing bases). The families first share a base at 4096, where the octal frame of order 24 and the hexagonal frame of order 12 both apply, and again at 2^{24} and 2^{36}. A box search restricted to six of twenty-four residues at base 4096 returned nothing; this is a null over a small slice and should not be read as absence.

Omega 5 (Constants by frame). The octal frame yields log 2, log 3, log 5; the hexagonal yields log 3 and log 7. A classification of which constants appear in which frame, indexed by the angles their defining inverse tangents require, has not been attempted. Theorem 5 suggests a constant is reachable in F_n only if expressible using angles that are multiples of 360/n.

Omega 6 (The winding coordinate). Theorem 2 shows the winding count is projected out. It is recoverable but unused. An extraction scheme sensitive to k rather than only to r would be a different class of construction.

§12 Master Theorem

Theorem 8 (Kernel-Frame-Winding-Residue). *Given a kernel of order n, an endpoint that is both a mark coordinate of F_n and a half-integer power of an integer base, and the requirement of positional digit extraction, the following are determined without further choice: the frame, as the* *kernel zero set; the shell denominators, one per mark indexed by winding; the shell index, one more than the numerator power; the coefficient weights, numerator coefficients scaled by powers of the endpoint; the algebraic support of the numerator, by the parity of the shell index; the orbit topology of each shell, by the parity of its modulus; the necessity of infinitely many windings, since refinement destroys compatibility while winding preserves it; and the status of the coefficient vector, as the shortest representative of an equivalence class modulo the null lattice of the shell constants.*

Within the class swept, exactly two endpoints satisfy the hypothesis: cos 45 and cos 60 degrees. The minimal frames containing them are F_8 at base 16 and F_6 at base 64, and both carry pi relations.

The integers appearing throughout are not magnitudes. The quotient records completed windings and is projected away; the residue selects an orientation. They carry traversal state rather than value. The formula is not a computation performed on numbers -- it is a reading taken from a frame that was closed before the first term was written, expressed in a coordinate that is itself only defined up to a direction the frame cannot see.

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